

Subject : Internet of Things

Module wise Question Bank

MODULE 1 : Introduction to Internet of Things (IoT)	
1.	Explain IoTWF Standardized Architecture with a neat diagram.
2.	Explain the concept of Edge computing, Fog computing and Cloud computing. How they are related to each other?
3.	What is IoT? How IoT is different from digitization? List out the different IoT challenges.
4.	Give the function of each layer of a seven-layer IoT architectural reference model published by IoTWF architectural committee.
5.	Explain in detail IoT impact.
6.	Explain in detail convergence of IT and OT.
7.	Discuss M2M IoT standardized architecture.
8.	What is IoT? Explain IoT blocks in detail.
9.	what is Fog computing? Give advantages and disadvantages of Fog computing.
10.	what is Edge computing? Give advantages and disadvantages of Edge computing.

MODULE 2 : Things in IoT	
1.	What is the smart object? Give the security and privacy concerns of smart object in IoT.
2.	Explain characteristics of smart object Give the trend in smart object.
3.	Describe Zigbee protocol stack using IEEE 802.15.4
4.	Define sensors in IoT Give classification of sensor and explain any four type.
5.	Explain compare and contrast wired and wireless sensor networks. Explain different topologies for WSN.
6.	Write a short note on MEMS.
7.	Explain following access technologies with application area of each 1) Zigbee 2) BLE
8.	Discuss clustered architecture of wireless sensor network.
9.	Differentiate between sensors and actuators with neat diagram.
10.	Explain the different types of sensors are used for measuring one of the physical properties and give its representative examples.

MODULE 3 : The Core IoT Functional Stack

1.	Explain in detail about Smart services in IoT system.
2.	Write short note on Data Analytics Vs Business Benefits.
3.	Discuss in brief- 1. Gateways and Backhaul Sub layer in Core IoT functional Stack 2. Communications Network layer in Core IoT functional Stack.
4	Give Classification of networks according to access technologies and distances.
5	Explain IoT application layer with neat diagram.
6	Explain in detail sensors and actuator layer.
7	Write short note on 1. Communication network layer. 2. Access network sublayer. 3. Network Transport layer. 4. IoT Network Management Sublayer
8	Discuss On 1.Application and Analytics layer 2. Analytics Vs control application. 3. Data Vs Network Analytics
9.	Explain the Smart Things:-architectural classification considering Layer I Thing sensor and actuator level.

MODULE 4 : Application Protocols for IoT

1.	What are the factors based on the type of device involved and the function it will performed helps to choose right protocol for a particular IOT application.
2.	Compare the suitable parameters COAP and MQTT application protocols used for IOT applications.
3.	Draw and Explain neat diagram of protocol stack four Transporting serial DNP3 SCADA over IP. Give meaning of a master slave relationship in DNP3.
4	Compare and contrast: Application layer protocol.
5	Explain IOT Application transport methods in brief.
6	Write short note on SCADA. Explain adapting SCADA for IP.
7	Explain MQTT. Compare COAP and MQTT.
8	Explain Generic-Web based protocol.
9.	Discuss IOT protocols.
10.	Explain network layer protocol.

MODULE 5 : Domain Specific IoTs	
1.	Describe 10 applications of IOT in existing market place. (all IoT domain)
2.	Explain in detail at least five case where IOT involvements will convert cities into smart cities.
3.	Describe top 10 applications of IOT in existing market place.
4.	Elaborate working model of Smart city.
5.	Explain health & lifestyle domain specific IoT.
6.	Explain different energy related IOT Domain with example.
7.	Describe various Home automation related IOT's domain with example.
8.	Describe - a) Retail b) Logistics c) Agriculture d) Environment e) Industry
9.	Describe domain specific IOT related to smart city.

MODULE 6 : Create your own IoT	
1.	Compare with suitable parameters between Raspberry Pi and Arduino.
2.	Why RESTful Json is popular choice for IOT application.
3.	Compare any five IOT software platform with suitable parameters.
4.	Explain features of ESP32
5.	Short note on JSON-LD
6.	Compare different IOT Boards in terms of connectivity
7.	Explain on IOT software platform - REST
8.	Explain Arduino Uno-board with it's pins and parts in detail.
9.	What are IOT software platform? Explain any five with example.
10.	Explain Raspberry Pi with diagram.
11.	Explain Amazon web services for IOT.
12.	Logical Design of IOT (IOT functional blocks)
13.	IOT Communication API in detail - REST-based